



Material List

Our recommendations to bring the research pickup point to life

In this document, we share our recommendations for the materials, devices, and setup needed to turn our concept into a real, working installation. Based on our research and testing, we explain what works best and what to consider when building it.

Introduction

Objective

This document explains what is needed to build and set up the installation. It focuses on practical choices and recommendations, rather than specific brands or suppliers. The goal is to show how this concept can realistically be put into practice, while keeping it flexible and easy to adapt for different events, spaces, or future updates.

Outline

- Main Display Setup
- Hologram Setup
- Optional Assets



Main Display Setup

The main screen is the heart of the experience. This is where visitors explore the projects and interact with the content.

Type of screen

We recommend using a **13-inch iPad Pro**. This is the largest iPad available and offers a high-quality display with smooth touch interaction. The 13-inch screen is large enough to be clearly visible, even for people standing slightly behind someone else. The app was designed specifically for this screen size, so everything fits and scales as intended.

Orientation

The screen is used in **landscape mode**. This allows more horizontal space for visuals and text, and works best with the layout of the interface and the project visuals.

Placement & Accessibility

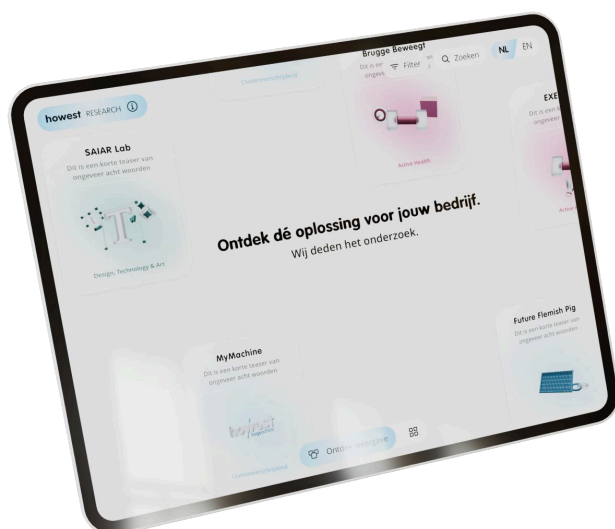
The iPad should be placed at a slight angle, either:

- mounted in a stand, or
- integrated directly into the hologram structure.

For good accessibility, the screen should be placed on a standing table, allowing visitors to interact with it easily while standing.

Installation of the App

The application can be installed directly on the device. You can find the exact instructions in the Readme file in the Github.



Hologram Setup

For the hologram, we researched several existing techniques, including Pepper's Ghost holograms, LED fan holograms, and volumetric displays. After comparing these options, we recommend using a **Pepper's Ghost hologram** with a single plexiglass plate. This technique offers the best balance between visual quality, cost, and practicality. It works well with screen-based content and allows for a large and clear hologram.

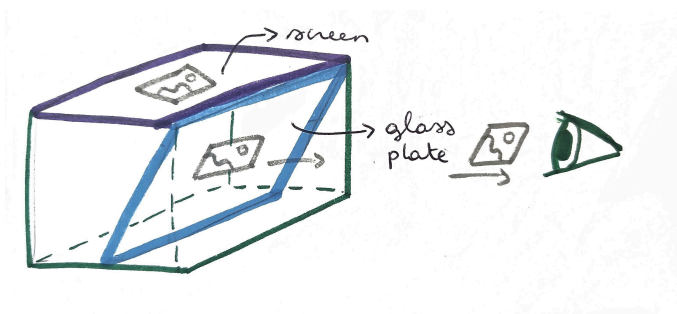


Technique: Pepper's Ghost (Single Plate)

We recommend using one plexiglass or acrylic plate rather than a multi-sided setup. This approach:

- Keeps the construction simple and clean
- Allows the hologram to be as large as the screen
- Works with regular video output (no special multi-angle content needed)
- Matches the way our application displays and controls the hologram

Multi-sided setups were considered but often result in smaller visuals, more visible glass, and extra complexity.



Screen for the Hologram

The Pepper's Ghost setup requires a second screen (hidden at the top of the box) to reflect the image onto the acrylic plate. This can be:

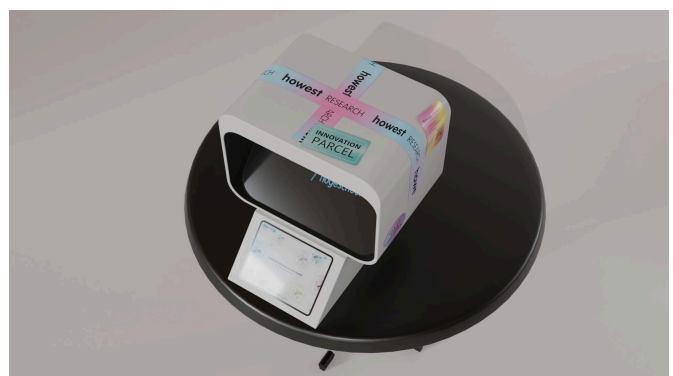
- A monitor connected to the iPad via HDMI or USB-C. (We used a regular monitor with an HDMI connection)
- A wireless display using Apple TV or any AirPlay-compatible device.

The quality and brightness of the screen will also be reflected onto the plexiglass plate.

Construction Recommendations

We recommend building the hologram box based on the visuals we provided, with the following considerations:

- Colour the inside of the box black to improve contrast and visibility.
- Leave the back open, as well as add some extra space in the back for a stronger floating effect.
- If needed, make an attachable back to improve visibility in bright environments.
- Angle the glass plate less than 45° to avoid the image looking projected onto the back.
- Integrate the screen directly into the hologram box design, making the setup feel more seamless and cohesive.
- To communicate the concept of a 'pick up point', add stickers on the box to make it look more like a parcel.
- Adjust the box dimensions to match the chosen screen size.



Optional Assets

In addition to the discovery screen and hologram (=the innovation parcel), we've designed extra assets that complement the concept and help communicate the "pickup" idea while attracting more attention.

Extra Parcels

We created five additional parcels (boxes) that can be added to the stand if space allows. Each represents a different domain of Howest Research and is styled to resemble a delivery box, reinforcing the overall theme.



Pick Up Sign

We also designed a pickup sign as an example of how this could look. The sign can be made to fold or collapse, making it easy to transport and save space.

